AQUA-SENSE: Ai-Powered Iot System For Public Water Quality Monitoring & Predictive Alerts

1. Hardware Requirements

- ESP32
- pH Sensor
- Turbidity Sensor
- TDS Sensor
- Temperature Sensor
- Gateway / Base Station
- LoRa Transceiver (SX1278)
- LoRa Receiver Module + ESP32
- 12V DC Adapter – Power supply

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2. Software Requirements

Frontend

- Flutter Framework
- Dart

Backend / Cloud

- Firebase Realtime Database
- Flask
- Google Colab / Jupyter Notebook

AI / Machine Learning

- TensorFlow / Keras
- NumPy
- Pandas
- Scikit-learn

Data Visualization

- Matplotlib
- Seaborn

Database / Cloud Storage

- Firebase Cloud Services

Programming Languages

- C / C++
- Python
- Dart

Development Tools & Platforms

- Arduino IDE
- GitHub
- Google Colab / Jupyter Notebook

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3. Required Development Packages & Libraries

- **ESP32 Board Package** – Required for ESP32 development using Arduino IDE
- **Flutter / Dart SDK** – Required for mobile application development
- **Python** – Required for AI, data processing, and backend development
- **Flask** – Python-based backend framework
- **Firebase** – Cloud database and real-time data storage
- **Required Python Libraries** – TensorFlow/Keras, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, and other project-specific dependencies
- **Required Flutter/Dart Libraries** – Dependencies required for Firebase integration, data visualization, API communication, and mobile application functionality